MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

FOURTH SEMESTER EXAMINATION

Model Question paper

MG4DSCCHE201-PHYSICAL CHEMISTRY I

Duration: 3 hrs. Maximum marks: 35

Session A (Select one question)

1. Study the variation of miscibility temperature of phenol-water system on addition of KCl. Use the data to determine the concentration of the given solution of KCl.
2. Determine the mass of given salt hydrate from transition temperature. You are provided with a salt hydrate of K_f = …… and molar mass of solute = ……
3. Find the molecular mass of the given solute by Rast's method. You are provided with a solvent of known mass and K_f = ……...
4. Construct the phase diagram of Naphthalene-Biphenyl system by plotting freezing point of different composition and determine the eutectic composition and eutectic temperature. Find out the composition of the given unknown mixture.

20 Marks

Session B

5. Lab report:

5 marks

Session C

6. Viva:

10 marks